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BIT 2114 Object Oriented Analysis and Design

LAST REVISION ON April 28, 2014





This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 4

Requirements Capture

Learning outcomes

Upon completing this topic, you should be able to:

1. Explain the importance of requirements capture.
2. Outline challenges faced in requirements capture.
3. Outline the sources of requirements.
4. Describe and carry out the elicitation process.



4.1. Requirements Capture

Part of the job of the system analysts is to find out from the users what they require in an information system. Indeed, identifying what a new system should be able to do is one of the first steps in development. The aim of developing a new information system must be to produce something that meets the needs of the people who will be using it.

Goguen defines requirements as properties that a system should have in order to succeed in the environment in which it will be used.

Sommerville and Sawyer define requirements as descriptions of how the system should behave, or of a system's property or attribute. Requirements are also the knowledge of the application domain and constraints on the operation of the system.



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Software projects are all too often undertaken without having a good understanding of the needs or wants of the client. Often projects fail because users' requirements are not accurately defined.

Boehm and Davis state that, the requirements phase of a large system development project is the most error prone, and these errors are the most expensive to correct. Consequently, this phase has the greatest economic leverage. Further, if the requirements are wrong, Sommerville says that the system may be delivered late and the cost more than originally expected. He adds that if the system continues its use, the cost of maintaining and evolving the system are very high. A basic question in Requirements Engineering is to find out what users really need.

As discussed in the previous chapter, many large projects fail



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because of inadequate requirements gathering. Requirements capture involves determining all of the sources of requirements and then using these to understand the problem to be solved and the characteristics of the solution. Correct requirements build quality into the product from the start. Attempting to test quality into code at the end of a project is expensive and ineffective. The probability that, defects originating from defective requirements will be completely redefined, decreases as they are allowed to propagate into the design code.

Good requirements promote agreement among developers and users, on the job to be done and the acceptance criteria for the delivered system. Developing requirements also provides a sound basis for resource estimation (cost, personnel quantity and skills, equipment, time). Improving one's organization com-



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petence in requirements capture will apply limited resources at the point of leverage to increase customer satisfaction and development team productivity. Brooks states that; 'The hardest single part of building a software system is deciding precisely what to build.

No other part of the conceptual work is as difficult as establishing the detailed technical requirements, including all the interfaces to people, to machines, and to other software systems. No other part of the work so cripples the resulting system if done wrong. No other part is more difficult to rectify late'

Collecting requirements is not an easy task because of several reasons:

1. A large number of requirements must be taken into account. For scheduling problems that have many constraints



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and may also have several stakeholders, several requirements normally arise.

2. Because of different stakeholders and different needs, several requirements are bound to arise which contradict each other, and need to be prioritized.
3. Requirements are inherently imprecise and often incomplete.
4. Implications of requirements and feasibility are hard to predict.
5. Different stakeholders with different backgrounds, interests and views are involved in the requirements development process.
6. Requirements change constantly.



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In as much as it is a challenging task developing requirements for a system, this does not compare with the cost of not developing requirements. For example, lack of proper requirements development was not the only cause, but it was a substantial contributor to the failure of the London ambulance service computer aided dispatch system.

4.2. Sources of Requirements

1. Stakeholders

A system stakeholder is anyone who benefits in a direct or indirect way from the system that is being developed (Sommerville and Sawyer, 2004). That is, stakeholders are those people or groups that will in some way, be affected by the system.

1. Application Domain



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The application domain places domain constraints, which generate system requirements and may place limitations on other requirements. The timetabling problem is one that has been researched upon numerous and some general constraints are expected of class timetabling systems.

1. The existing system

The need for an automated system could be born from the fact that currently a manual system is being used and a computer is used only for typing and printing. Also, there could be a need to replace an old computer system with latest technology. The existing system has to be understood so as to facilitate developing requirements for a new computerized system. It also aids in understanding why the manual system is being used and why there is need for a computerized system. The existing system



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also provides documents that are examined and are used as a source of requirements.

4.2.1. Stakeholder Identification

As part of the requirements capture process, all potential stakeholders should be explicitly defined and consulted to discover if they have specific requirements or if they impose specific constraints on the system.

The reasons for careful consultation of stakeholders include:

1. If everyone who is likely to be affected by the system is not considered, important requirements are likely to be missed and the system may have to be reworked to include these at a later stage or, in the worst case, may not be used.
2. Identification of stakeholders can help in discovery of view-



points for structuring the system requirements. This is especially important if a viewpoint oriented approach to structure the requirements elicitation process is used. Viewpoints are a way of structuring the requirements to represent the perspective of different stakeholders.

3. Identifying stakeholders and discussing the system with them, makes them feel that they are part of the development process. They are more likely to be sympathetic to the introduction of the system and to volunteer information about their requirements.

Sommerville and Sawyer (2004), suggested that the following checklist be used in identifying stakeholders:

1. By discovering the potential end-users of the system and the customers of the organization who will use the system.



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Determining the needs of users will involve understanding the activities certain users are performing in order to accomplish their work (Jirotko and Lincoln, 2000).

2. By initial discussion with organizational management and by considering descriptions of the business processes that the system is intended to support, and the people involved.
3. By considering engineers and maintenance staff responsible for developing and maintaining the system.

Example Stakeholders:

- Users: concerned with the features and functionality of the new system
- Designers: want to build a perfect system, or reuse existing code



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- Systems analysts : want to “get the requirements right”
- Training and user support staff: want to make sure the new system is usable and manageable
- Business analysts: want to make sure “we are doing better than the competition”
- Technical authors: will prepare user manuals and other documentation for the new system
- The project manager: wants to complete the project on time, within budget, with all objectives met.
- “The customer”: Wants to get best value for money invested!



4.2.2. The Elicitation Process

The elicitation process focuses on understanding the users' needs and how to communicate these effectively. There are several elicitation techniques available and the general consensus is that there is no 'best' one.

Introspection

Introspection involves imagining what kind of system a user would want if they were to use an automated or new system. Introspection is a useful method but since it's biased towards one's personal opinion, its results cannot be entirely relied upon and need to be used in conjunction with the results from the other techniques.



Interviews

Open-ended and/or unstructured interviews can be used; Open-ended interviews are more likely to obtain unrehearsed and spontaneous information, while unstructured interviews allow the interviewee to be more at ease and talk informally about the questions asked.

Questionnaires

Questionnaires can be used to elicit requirements in situations where the anonymity of the respondents is required or just to gain more information from the users. Participation/Observation This involves observing the way users interact with an existing system and also taking part in its use and/or creation.



Prototyping

A simple throwaway prototype can be used to elicit additional requirements. The prototype may or may not employ an algorithm to implement the system, but rather simply provides a possible graphical user interface of the proposed system.

4.2.3. Starting Points for Elicitation

1. Boundaries
 - Scoping the problem
2. Stakeholders
 - Identifying the problem owners
3. Goals
 - Identifying the success criteria



4. Scenarios

- Using concrete examples to understand the problem

4.2.4. Where do we Start?

1. Identify the problem

- what is the objective of the project?
- the “vision” of those who are pushing for it? e.g., “Scheduling meetings is too costly right now”

1. Scope the problem

- given the vision, how much do we tackle? e.g. “Build a system that schedules meetings”, ...or... e.g. “Build a system that maintains people’s calendars” ...or...

2. Identify solution scenarios



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- given the problem, what is the appropriate business process for solving it? e.g. “Anyone who wants to schedule a meeting goes to the secretary, gives details and the secretary handles the rest”, ...or...
3. Scope the solution
- Given a business process, what parts should be automated, and how? e.g. “Computer takes in scheduling request details, outputs a solution” ...or... e.g. “Solution arrived at interactively by secretary and computer” ...or...

4.2.5. Starting Point

1. Some notion that there is a “problem” that needs solving
 - e.g. dissatisfaction with the current state of affairs



- e.g. a new business opportunity
- e.g. a potential saving of cost, time, resource usage, etc.

2. Collect enough information to:

- identify the “problem”/”opportunity”
 - Which problem needs to be solved? (identify problem Boundaries)
 - Where is the problem? (understand the Context/Problem Domain)
 - Whose problem is it? (identify Stakeholders)
 - Why does it need solving? (identify the stakeholders’ Goals)
 - How does the problem manifest itself? (collect some Scenarios)



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- When does it need solving? (identify Development Constraints)
- What might prevent us solving it? (identify Feasibility and Risk)
- Become an expert in the problem domain
- Learn how to find your way round a new problem area quickly
- Use your (initial) ignorance as an excuse to ask (dumb?) questions
- Recognize the domain expertise of the people you talk to

3. Vague problem stated by the customer:

- E.g. university textbook store:
 - Manager wants to computerize the book order forms



filled out by instructors;

- E.g. A large insurance company:
 - Claims manager wants to cut down the average time it takes to process an insurance claim from 2 months to 2 weeks
- E.g. A telecommunications company:
 - CIO wants to integrate the billing system with customer record systems of several affiliates, so there is only one billing system...
- E.g. Large Government Aerospace Agency:
 - The president wants to send a manned mission to Mars by the the year 2020

1. Often you only see symptoms rather than causes:



- E.g. “JKUAT patients needing X-ray scans have to wait for months”
 - The long wait is the symptom, not the problem. The problem may be:
 - * Shortage of X-ray machines;
 - * Shortage of trained staff;
 - * Shortage of doctors to process the data
 - * Inefficient scheduling procedures

4.3. Problems Faced While Eliciting Requirements.

A good requirements elicitation process supports the development of a specification that is unambiguous, complete, verifiable, consistent, modifiable, traceable, and usable, to name but a few. However, problems with requirements elicitation inhibit



the definition of such requirements.

1. Stakeholders often do not really know what they want from the computer system except in the most general terms. Even when they have a clear idea of what they would like the system to do, they often find this difficult to articulate. They may make unrealistic demands because they are unaware of the costs of their requests. This was dealt with by eliciting information about the tasks users currently perform and those that they may want to perform. And also recognizing the users as domain experts and employing different elicitation techniques.
2. The 'yes, but' syndrome. It is human nature to either like something very much or to like something but to find faults in it or to suggest 'what could be better'. This problem



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was dealt with by employing techniques that bring out the 'yes, buts' early enough and also plan for time to get feedback from users.

3. The requirements engineer may not know when to stop eliciting for requirements. This was dealt with by scoping the requirements effort and defining the problems that are to be solved by the system.
4. The stakeholders may not understand terms that the requirements engineer is using and may give answers not required. For example, from the interview with the timetabling master, when asked about constraints, the timetabling master spoke of the problems they faced.
5. The large amount of information gathered during elicitation necessitates that it be structured in some way. The



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understanding of this structure is dependent on the characteristics of the stakeholder communities.

6. The language used to express the requirements back to the stakeholder communities may be too formal or too informal to meet the needs of each of the groups.
7. Requirements often change. During the time it takes to develop a system the users' needs may mature because of increased knowledge brought on by the development activities, or they may shift to a new set of needs because of unforeseen organizational or environmental pressure. Requirements may emerge from new stakeholders who were not originally consulted. Additionally, the economic and business environment in which the analysis takes place is dynamic. It inevitably changes during the elicitation pro-



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cess. Hence the importance of particular requirements may change.

8. Some users did not respond to the questionnaires.

A major goal in Requirements Engineering is to minimize such problems. This has involved putting significant effort into requirements elicitation. This process is not a one-off procedure but an iterative process that goes on throughout the requirements gathering process.



Learning Activities

Carry out requirements capture in the organization that you are working in. This assignment has ten steps as outlined below:

1. Identify a problem within the organization you are working with that is amenable to requirements engineering techniques.
2. Scope the problem, i.e., choose how small or large a problem you will tackle during your feasibility study.
3. Interview key people involved in the problem. These may include your contacts, plus others with responsibilities related to the feasibility study.
4. Study relevant documents. This may involve reading policy documents, memos, documentation on current systems



and new objectives/needs.

5. Define alternatives for conducting business and for automation. Define the criteria you will use to evaluate and choose among these alternatives. Your criteria should include cost/benefit, where applicable.
6. Create models of the key aspects of the problem and the alternative solutions. These should include:
 - (a) Goal models to represent the relationships between goals of the key stakeholders;
 - (b) Activity diagrams and/or business process models to represent important business processes related to the problem and the alternative solutions.
7. Assess unusual circumstances or special attention items.



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This may involve special requirements for particular employees or customers of the organization.

8. Evaluate the alternatives on technical, economic, schedule and operational feasibility, and choose the one that looks most promising.
9. Write a report that describes the objectives of your study, the problem you identified the alternatives you explored, as well as your recommendations.

Revision Questions

Example . Briefly describe the phases of the UML process

Solution: for revision



EXERCISE 1.  State the normal procedure for identifying the



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following, from a textual description. (a) Objects (b) Operations

EXERCISE 2.  Define the term UML, explaining its difference from other types of Object Modeling Techniques (OMTs)



References and Additional Reading Materials

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